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MID SEMESTER EXAMINATION - 2025

Name of the Course: **B. Tech. Petroleum Engineering**
 Name of the Paper: **Advanced Mathematics**
 Time: **1and 1/2 Hours**

Semester: **Third**
 Paper Code: **TMA-307**
 Maximum Marks: **50**

Note: -

- (i) All questions are compulsory.
- (ii) Answer any one sub questions among a & b in each main question.
- (iii) Each question carries 10 marks.

Q1	(10marks)	
(a)	Write the general forms of the following partial differential equations: (i) One dimensional wave equation. (ii) One dimensional heat equation. (iii) Laplace equation. Also, explain the boundary value problem.	CO1
	OR	
(b)	Initially a tightly stretched string with fixed ends $x = 0$ and $x = l$ is in a position given by $y(x, 0) = y_0 \sin^3 \frac{\pi x}{l}$. If it is released from rest from this position, find the displacement $y(x, t)$	
Q2	(10 marks)	
(a)	Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with the boundary condition $u(x, 0) = 3 \sin n\pi x, u(0, t) = 0, u(l, t) = 0$, where $0 < x < l$	CO1
	OR	
(b)	A rod of length l with insulated sides is initially at a uniform temperature u_0 . Its ends are suddenly cooled to 0°C and are kept at that temperature. Find the temperature function $u(x, t)$	
Q3	(10 marks)	
(a)	In a bolt factory machines, A, B, C manufacture respectively 25%, 35% and 40% of the total. Of their output 5%, 4% and 2% are defective. A bolt is drawn at random from the product and found to be defective. What are the probabilities that it was drawn at random from the (i) machine A (ii) machine B (iii) machine C.	CO1
	OR	

(b)	A die is rolled 10 times. Find the probability that 2 appears exactly 9 times.						
Q4	(10 marks)						
(a)	Find the mean and median of the data:						CO4
	Annual sales (in thousands)	10-20	20-30	30-40	40-50	50-60	
	Number of companies	6	17	17	22	8	
	OR						
(b)	Find the Karl Pearson's coefficient of skewness of the marks obtained by the students in a college.						
	Marks obtained	0-10	10-20	20-30	30-40		
	No. of students	10	12	18	25		
Q5	(10 marks)						
(a)	Find the variance of the given data						CO4
	Investment (lakhs rupees)	100-200	200-300	300-400	400-500	500-600	
	Number of Investors	18	12	27	25	21	
	OR						
(b)	On an average 3% of the electric bulbs manufactured by a company are defective. Using Poisson's distribution, find the probability that in a sample of 100 bulbs (a) There are 5% defective bulbs. (b) No defective bulb.						